Direct Drive of a Buck Converter by Delta-Sigma Modulation at 13.56-MHz Sampling

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***Abstract*—A direct drive of a buck converter is proposed in this paper. It is achieved by delta-sigma modulation at 13.56-MHz sampling without digital PWM. The high-speed gate driver, which is based on GaN-HEMTs push-pull configuration, is employed to drive a SiC power MOSFET at high switching frequency. By delta-sigma modulation at 13.56-MHz sampling, the output voltage of buck converter is directly controlled in the wide range.**

Keywords—Buck converter; Delta sigma modulation; High-speed gate driver; SiC MOSFET

I. Introduction

Switching frequency of power converters has been enhanced through recent progress of power switching devices [1]–[3]. High-frequency DC-DC converters were reported at over 1 MHz using SiC metal-oxide-semiconductor field-effect transistors (MOSFETs) [4], [5]. Recently, 10-MHz hard switching of a SiC MOSFET was reported driven by a high-speed gate driver using GaN high electron mobility transistors (HEMTs) [6]. The enhancement of switching frequency contributes to downsizing of power converters and leading advanced applications related to power processing. However, it is difficult to achieve the precise control of duty ratio for pulse width modulation (PWM) at high-frequency hard switching. Here, we propose a pulse density modulation (PDM) using a delta-sigma modulator to achieve precise control of high-frequency power converters.

Delta-sigma modulation is widely used for analog-digital (A-D) conversion, and also adopted to power converters owing to its noise shaping property [7]–[9]. A second-order delta-sigma modulator can strongly reduce noise from power converters [10]–[12]. For this purpose, digital PWM (DPWM) has been also employed with the delta-sigma modulator because signals from a delta-sigma modulator would be sparse at low switching frequency.

In this report, we try to directly drive a valve transistor with delta-sigma modulation by enhancing the sampling frequency up to 13.56 MHz. At the high sampling frequency, subsequent DPWM is not required and the driving configuration becomes simple. We investigate a buck converter with a SiC power MOSFET driven by delta-sigma modulation at the sampling frequency of 13.56 MHz, and achieve the wide range of voltage regulation.

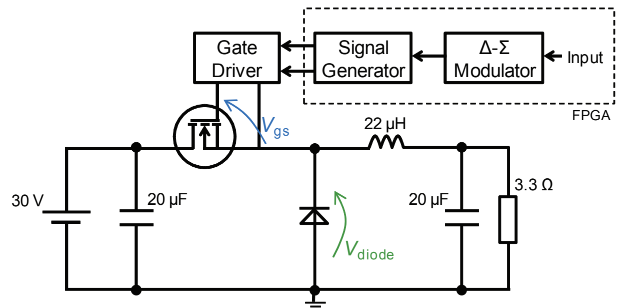

Fig. 1. Schematic of fabricated buck converter using delta-sigma modulator.

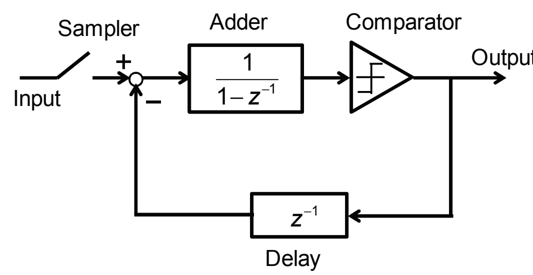

Fig. 2. Diagram of implemented delta-sigma modulator.

II. Proposed Architecture

The schematic of the fabricated buck converter is shown in Fig. 1. SiC MOSFET (ROHM SCT2450KE, 1200 V, 10 A) and SiC SBD (ROHM SCS208AGC, 650 V, 8 A) are employed. The inductor of the output stage is $22\,\mu$H and the output capacitor is $20\,\mu$F. A gate driver based on GaN-HEMT push-pull configuration is adopted to drive the SiC MOSFET [3]. In order to generate driving signals, a delta-sigma modulator is implemented in FPGA board (Xilinx, ZYBO Zynq-7000). The diagram of the implemented delta-sigma modulator is shown in Fig. 2. A first-order delta-sigma modulator is constructed in the FPGA board at sampling frequency of 13.56 MHz. A photograph of the fabricated buck converter and delta-sigma modulated gate driver are shown in Fig. 3. Power DC supplies (Matsusada, P4K18-2) are used for the gate driver and a regulated DC power supply (Takasago, KX-210L) is used for the input of the buck converter. Switching waveforms of the

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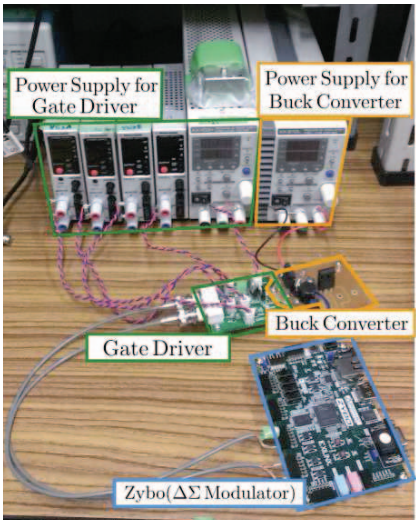

Fig. 3. A photograph of fabricated buck converter and its experimental configuration.

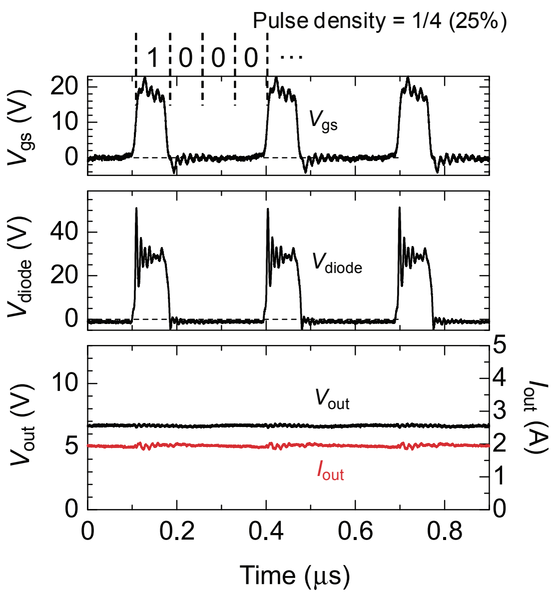

Fig. 4. Experimental switching waveforms of buck converter at pulse density 25%. The sampling frequency of delta-sigma modulation is set at 13.56 MHz. The input voltage is at 30 V.

buck converter are acquired with an oscilloscope (KEYSIGHT, InfiniiVision DSO-X 6004A).

III. Experimental Results

The experimental switching waveforms of the buck converter are shown in Fig. 4. The pulse density was set at 1/4, corresponding to the conventional duty ratio of 25% in PWM-based buck converter. The sampling frequency was set at 13.56 MHz. Depending on the high frequency characteristics

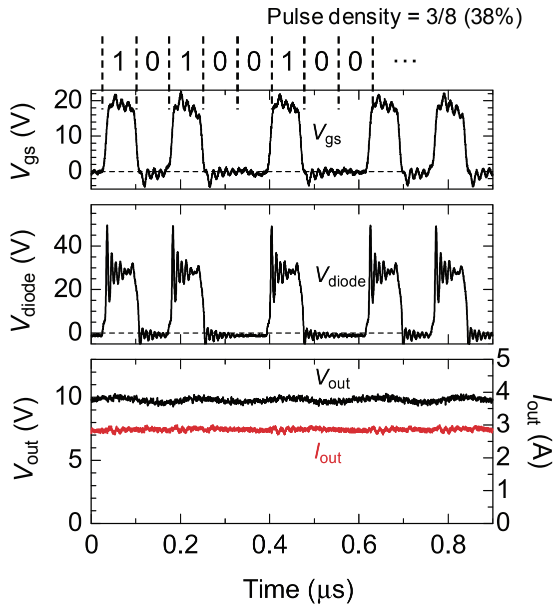

Fig. 5. Experimental switching waveforms of buck converter at pulse density 38%. The sampling frequency of delta-sigma modulation is set at 13.56 MHz. The input voltage is at 30 V.

of SiC MOSFET and the direct drive capability of the gate driver, clear switching waveforms of the gate-source voltage V_{gs} and the diode voltage V_{diode} were obtained even at 13.56-MHz sampling. As a result, the output voltage V_{out} was regulated at 6.8 V against the input voltage of 30 V.

Figure 5 shows the experimental switching waveforms of the buck converter at pulse density 38%. The pulse density of the obtained switching waveforms was increased compared to the experimental results at pulse density 25%, so that the output voltage increased to 10.5 V.

Figure 6 shows the measured output voltages of the buck converter with various pulse densities. The input voltage was 30 V. Varying the pulse density of the delta-sigma modulator, the output voltage was successfully controlled in the wide range. Owing to the high-frequency gate driver, pulse density modulation can be directly employed for the valve transistor to control the output voltage.

IV. Discussion

The ideal output voltage is shown in Fig. 6 by a dotted line, given by $V_{out} = \mu V_{in}$ where μ is the average density of the pulses. The output voltage was controlled in the wide range, but the measured output voltage is lower than the ideal output voltage, in particular, at high pulse density. This is because the conduction losses in the MOSFET, the SBD, and the inductor cannot be ignored with compared to the load resistance of $3.3\,\Omega$.

Here, we describe the calculation of output voltage by taking the conduction losses into account. The circuit equation at the on-state of the SiC MOSFET is given by

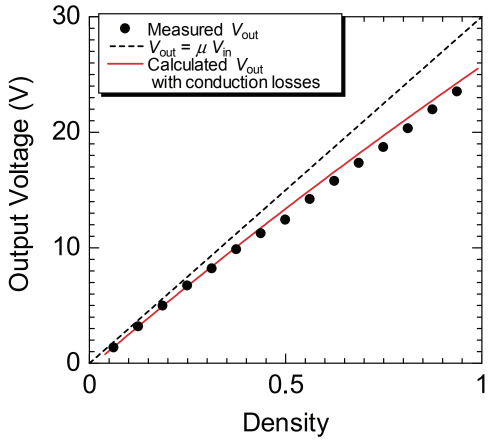

Fig. 6. Measured output voltages of buck converter due to pulse density of the delta-sigma modulator and estimated output voltage with conduction losses. The sampling frequency of the delta-sigma modulation is set at 13.56 MHz. The input voltage is at 30 V.

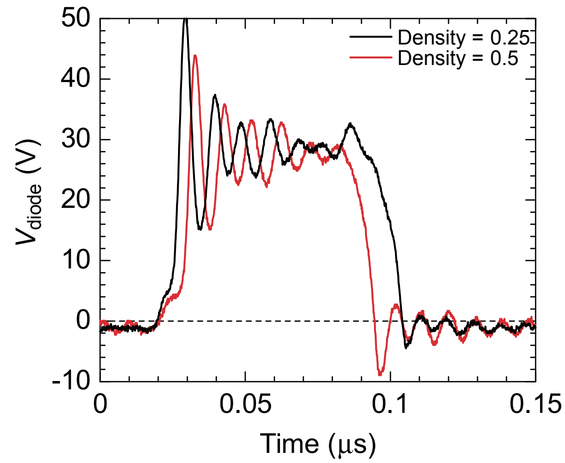

Fig. 7. Voltage waveforms of the SiC SBD at pulse densities of 0.25 and 0.5.

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\phi\\ q\end{bmatrix}=\begin{bmatrix}-(R_{\mathrm{mos}}+R_{\mathrm{L}}) & -1\\ 1 & -1/R_{\mathrm{load}}\end{bmatrix}\begin{bmatrix}I_{\mathrm{L}}\\ V_{\mathrm{out}}\end{bmatrix}+\begin{bmatrix}E\\ 0\end{bmatrix} \tag{1}$$

where ϕ denotes the magnetic flux at the inductor, q the charge at the capacitor, I_{L} the inductor current, R_{mos} the on-resistance of the SiC MOSFET, R_{L} the series resistance of the inductor, and R_{load} the load resistance, respectively. The circuit equation at the off-state of the SiC MOSFET is given by

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\phi\\ q\end{bmatrix}=\begin{bmatrix}-(R_{\mathrm{D}}+R_{\mathrm{L}}) & -1\\ 1 & -1/R_{\mathrm{load}}\end{bmatrix}\begin{bmatrix}I_{\mathrm{L}}\\ V_{\mathrm{out}}\end{bmatrix}+\begin{bmatrix}-V_{\mathrm{f}}\\ 0\end{bmatrix} \tag{2}$$

where V_{f} corresponds to the forward voltage drop at the SiC SBD, and R_{D} is introduced to describe the series resistance at the SiC SBD. Here, the current-voltage characteristics of the SiC SBD is approximated by $V = V_{\mathrm{f}} + R_{\mathrm{D}} \times I$. Using the method of state-space averaging for the above equations, the circuit equations are merged into the following state equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\phi\\ q\end{bmatrix}=\begin{bmatrix}-R_{\mathrm{cond}}(\mu) & -1\\ 1 & -1/R_{\mathrm{load}}\end{bmatrix}\begin{bmatrix}I_{\mathrm{L}}\\ V_{\mathrm{out}}\end{bmatrix}+\begin{bmatrix}\mu E-(1-\mu)V_{\mathrm{f}}\\ 0\end{bmatrix} \tag{3}$$

$$R_{\mathrm{cond}}(\mu)=\mu R_{\mathrm{mos}}+(1-\mu)R_{\mathrm{D}}+R_{\mathrm{L}} \tag{4}$$

using the notation of the pulse density μ. From this equation, the output voltage is estimated by

$$V_{\mathrm{out}}=\frac{\mu E-(1-\mu)V_{\mathrm{f}}}{1+R_{\mathrm{cond}}/R_{\mathrm{load}}}. \tag{5}$$

The on-resistance of the SiC MOSFET is $0.5\,\Omega$ obtained from the static characteristics, and the series resistance R_{D} and the forward voltage drop V_{f} of the SiC SBD are $0.11\,\Omega$ and 0.7 V, respectively. The series resistance of the inductor is $0.08\,\Omega$ measured with an impedance analyzer. The estimated output voltage is shown in Fig. 6 by a solid line. The estimated voltage well represents the measured output voltage. Hence, the voltage decrease at the high pulse density is caused by the conduction losses at the circuit elements.

Switching losses also decrease the output voltage. At the pulse density in the vicinity of 0.5 in Fig. 6, the measured output voltage is slightly lower than the estimated output voltage. In the pulse density modulation, the number of switching times approaches to the maximum at the pulse density 25%. Thus, the switching losses can affect the output voltage in this region.

The effective pulse width is important to estimate the effect of switching losses because the switching characteristics of the SiC MOSFET depends on the drain current. Figure 7 shows the voltage waveforms of the SiC SBD at pulse densities of 25% and 50%. The measured output current is 1.9 A at the pulse density 25% and 3.3 A at the pulse density 50%. At the higher current, the fall time of the SiC MOSFET decreased. As a result, the effective pulse width decreased in the pulse density modulation, resulting that the output voltage decreases at higher pulse density.

Besides, the effect of the switching losses on the output voltage is relatively small in this study compared to the conduction losses because of the fast switching characteristics of SiC MOSFET. In order to estimate the switching losses qualitatively, it is necessary to precisely model the dynamic characteristics of SiC MOSFET and develop a simulation environment for design of high-frequency converters.

V. Conclusion

The direct drive was investigated by delta-sigma modulation for DC-DC buck converter. Through the recent progress of power switching devices and gate drivers, the fast hard switching was obtained at 13.56 MHz. By using the gate driver with delta-sigma modulation, the output voltage was directly regulated without digital PWM. Concepts constructed in digital circuits should be rebuiled in high-frequency power converters with next-generation power devices.

Acknowledgment

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References

[1] H. Matsunami and T. Kimoto, "Step-Controlled Epitaxial Growth of SiC: High Quality Homoepitaxy," *Mater. Sci. Eng.*, vol. R20, no. 3, pp. 125–166, 1997.

[2] J. A. Cooper, Jr, M. R. Melloch, R. Singh, A. Agarwal, and J. W. Palmour, "Status and Prospects for SiC Power MOSFETs," *IEEE Trans. Electron Devices*, vol. 49, no. 4, pp. 658–664, Apr. 2002.

[3] T. Kimoto, "Material Science and Device Physics in SiC Technology for High-Voltage Power Devices," *Jpn. J. Appl. Phys.*, vol. 54, pp. 40103-1–40103-27, 2015.

[4] A. M. Abou-Alfotouch, A. V. Radun, H.-R. Chang, and C. Winterhalter, "A 1-MHz Hard-Switched Silicon Carbide DC-DC Converter," *IEEE Trans. Power Electronics*, vol. 21, no. 4, pp. 880-889, Jul. 2006.

[5] A. Kadavelugu, S. Beak, S. Dutta, S. Bhattacharya, M. Das, A. Agarwal, J. Scofield, "High-Frequency Design Considerations of Dual Active Bridge 1200 V SiC MOSFET DC-DC Converter," *2011 Twenty-Sixth Annual IEEE Applied Power Electronics Conference and Exposition*, pp. 314-320, 2011.

[6] K. Nagaoka, K. Chikamatsu, A. Yamaguchi, K. Nakahara, and T. Hikihara, "High-Speed Gate Drive Circuit for SiC MOSFET by GaN HEMT," *IEICE Electron. Express*, vol. 12, p. 20150285, 2015.

[7] A. Hirota, S. Nagai, and M. Nakaoka, "A Novel Delta-Sigma Modulated DC-DC Power Converter Utilizing Dither Signal," *2000 IEEE 31st Annual Power Electronics Specialists Conference*, vol. 2, pp. 831-836, 2000.

[8] H. Ahmad and B. Bakkaloglu, "A Digitally Controlled DC-DC Buck Converter Using Frequency Domain ADCs," *2010 Twenty-Fifth Annual IEEE Applied Power Electronics Conference and Exposition*, pp. 1871-1874, 2010.

[9] W. Yan, W. Li, and R. Liu, "A Noise-Shaped Buck DC-DC Converter With Improved Light-Load Efficiency and Fast Transient Response," *IEEE Trans. Power Electronics*, vol. 26, no. 12, pp. 3908-3924, Dec. 2011.

[10] O. Trescases, Z. Luki, W. T. Ng, and A. Prodic, "A Low-Power Mixed-Signal Current-Mode DC-DC Converter Using a One-Bit Δ-Σ DAC," *Twenty-First Annual IEEE Applied Power Electronics Conference and Exposition*, pp. 700-704, 2006.

[11] Z. Lukic, N. Rahman, and A. Prodic, "Multibit Σ-Δ PWM Digital Controller IC for DC-DC Converters Operating at Switching Frequencies beyond 10 MHz," *IEEE Trans. Power Electronics*, vol. 22, no. 5, pp. 1693-1707, Sep. 2007.

[12] B. Li, X. Lin-Shi, B. Allard, and J. M. Retif, "A Digital Dual-State-Variable Predictive Controller for High Switching Frequency Buck Converter with Improved Σ-Δ DPWM," *IEEE Trans. Industrial Informatics*, vol. 8, no. 3, pp. 472-481, Aug. 2012.